FIITJEE **JEE (Main)-2018**

ALL INDIA TEST SERIES

FULL TEST – VII

Time Allotted: 3 Hours **Maximum Marks: 360**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section-A (01 – 30, 31 – 60, 61 – 90)** *contains 90 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

SECTION – A

Straight Objective Type

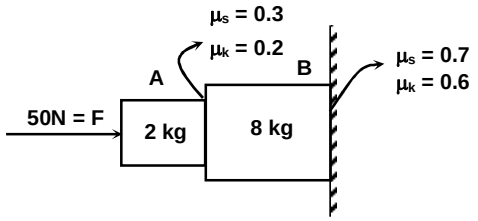
This section contains **30 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A particle moves along a circle of radius 'R' with speed varies as $v = a_0t$, where a_0 is a positive constant. Then the angle between the velocity vector and the acceleration vector of the particle when it has covered one fourth of the circle is

(A) $\tan^{-1}\left(\frac{\pi}{4}\right)$ (B) $\tan^{-1}\left(\frac{\pi}{2}\right)$

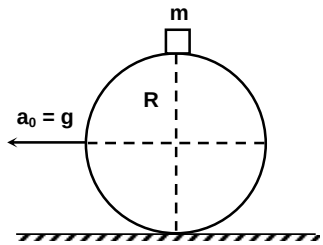
(C) $\tan^{-1}(\pi)$ (D) $\tan^{-1}(2\pi)$

2. Two blocks A and B of masses 2 kg and 8 kg respectively are kept in contact and pressed against a rough vertical wall by applying a horizontal force F = 50 N as shown. The frictional force acting between the two blocks is



(A) zero
(B) 6N
(C) 8 N
(D) 10 N

3. A small body of mass 'm' is placed at the top of a smooth sphere of radius 'R' placed on a horizontal surface as shown. Now the sphere is given a constant horizontal acceleration $a_0 = g$. The velocity of the body relative to the sphere at the moment when it loses contact with the sphere is

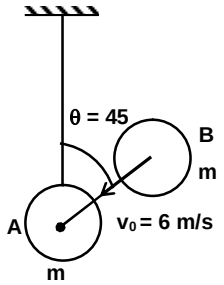


(A) $\sqrt{\frac{gR}{3}}$ (B) $\sqrt{\frac{2gR}{3}}$

(C) $\sqrt{\frac{5gR}{3}}$ (D) $\sqrt{\frac{7gR}{3}}$

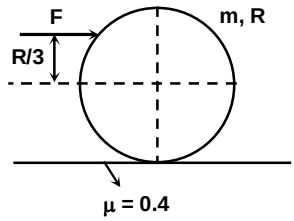
Space for rough work

4. A ball of mass m is attached to the lower end of a vertical string whose upper end is fixed. An identical ball 'B' moving with a velocity v_0 = 6 m/s at an angle θ = 45° from vertical collides elastically with the ball 'A' as shown. The velocity of the ball 'B' just after collision is



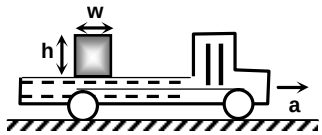
(A) 5 m/s
(B) 4 m/s
(C) 3 m/s
(D) 2 m/s

5. A thin uniform spherical shell of mass m = 2 kg and radius R = 10 cm is placed on a rough horizontal surface with a coefficient of friction μ = 0.4. A horizontal force 'F' is applied to the spherical shell at a height R/3 above the centre as shown. The maximum magnitude of the applied force 'F' for which the spherical shell will be rolling without slipping is (take g = 10 m/s^2)



(A) 20 N (B) 30 N
(C) 40 N (D) 60 N

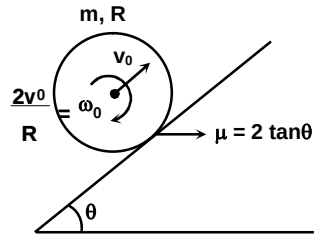
6. A block of mass 2 kg is kept on a truck moving with a constant acceleration. The height 'h' of the block is 60 cm and its width 'w' is 15 cm. The surface between the block and the truck is sufficiently rough to prevent slipping. The minimum acceleration of the truck at which the block will topple is (take g = 10 m/s^2)



(A) 2.5 m/s^2 (B) 3.5 m/s^2
(C) 4.5 m/s^2 (D) 5.5 m/s^2

Space for rough work

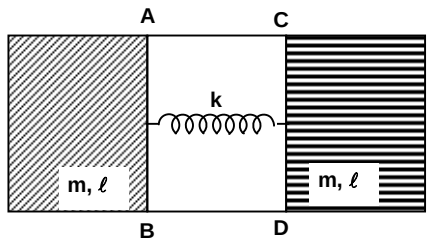
7. A uniform solid sphere of mass m and radius 'R' is imparted an initial velocity v_0 and angular velocity $\omega_0 = \frac{2v_0}{R}$ and then placed on a rough inclined plane of inclination 'θ' and coefficient of friction $\mu = 2\tan\theta$ as shown. The time after which the sphere will start rolling without slipping is



(A) $\frac{v_0}{2g\sin\theta}$ (B) $\frac{v_0}{4g\sin\theta}$

(C) $\frac{v_0}{6g\sin\theta}$ (D) $\frac{v_0}{8g\sin\theta}$

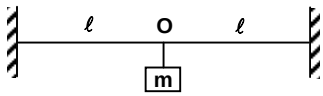
8. Two thin films of same liquid of surface tension 'T' are formed between a smooth rectangular wire frame and two thin uniform straight wires each of mass 'm' and length 'l' connected to a massless non-deformed spring of stiffness 'k' and then the system is released from rest. The maximum elongation produced in the spring is



(A) $\frac{2T\ell}{K}$ (B) $\frac{4T\ell}{K}$

(C) $\frac{6T\ell}{K}$ (D) $\frac{8T\ell}{K}$

9. A steel wire of negligible mass, length 2l, cross sectional area 'A' and Young's modulus 'Y' is held between two rigid walls. A small body of mass 'm' is suspended from its middle point 'O' as shown. The vertical descent of the middle point 'O' of the wire is

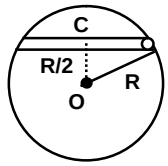


(A) $\ell\left(\frac{mg}{AY}\right)^{1/3}$ (B) $\ell\left(\frac{2mg}{AY}\right)^{1/3}$

(C) $\ell\left(\frac{4mg}{AY}\right)^{1/3}$ (D) $\ell\left(\frac{6mg}{AY}\right)^{1/3}$

Space for rough work

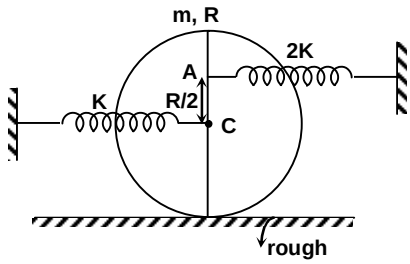
10. A tunnel is dug across the earth of mass 'M' and radius 'R' at a distance R/2 from the centre 'O' of the Earth as shown. A body is released from rest from one end of the smooth tunnel. The velocity acquired by the body when it reaches the centre 'C' of the tunnel is



(A) $\sqrt{\frac{3GM}{2R}}$ (B) $\sqrt{\frac{3GM}{4R}}$

(C) $\sqrt{\frac{3GM}{8R}}$ (D) $\sqrt{\frac{3GM}{16R}}$

11. A uniform circular disc of mass 'm' and radius 'R' is placed on a rough horizontal surface and connected to the two ideal non-deformed springs of stiffness k and 2k at the centre 'C' and point 'A' as shown. The centre of the disc is slightly displaced horizontally from equilibrium position and then released, then the time period of small oscillation of the disc is (there is no slipping between the disc and the surface).



(A) $2\pi\sqrt{\frac{5m}{7k}}$ (B) $2\pi\sqrt{\frac{3m}{7k}}$

(C) $2\pi\sqrt{\frac{5m}{11k}}$ (D) $2\pi\sqrt{\frac{3m}{11k}}$

12. Two moles of an ideal monoatomic gas undergoes a process VT = constant. If temperature of the gas is increased by ΔT = 300 K, then the ratio $\left(\frac{\Delta U}{\Delta Q}\right)$ is

(A) 2 (B) 3

(C) 4 (D) 6

13. A heat engine employing a carnot cycle with an efficiency of η = 10% is used as a refrigerating machine with the same thermal reservoirs. Then its coefficient of performance is

(A) 3 (B) 5

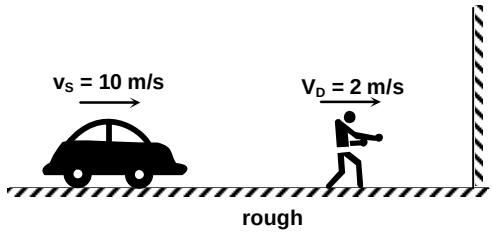
(C) 7 (D) 9

Space for rough work

14. A string of length 50 cm and mass 2.5 g is fixed at both ends. A pipe closed at one end has a length of 85 cm. When the string vibrates in its second overtone and the air column in the pipe in its first overtone, they produce a beats frequency of 6 Hz. It is also observed that decreasing the tension in the string decreases beats frequency. Neglect the end correction in the pipe and velocity of sound in air is 340 m/s. then the tension in the string is

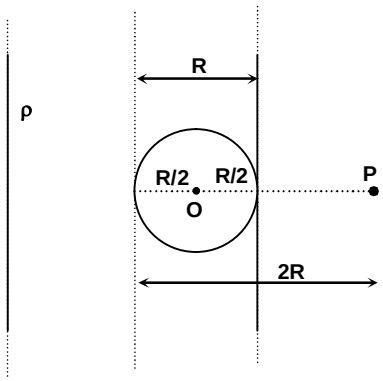
(A) 72 N (B) 52 N
(C) 26 N (D) 18 N

15. A car is moving with a constant velocity 10 m/s towards a stationary wall. An observer in between the car and the wall is moving with a constant velocity 2 m/s towards the stationary wall. The car blows a horn of frequency 320 Hz. The velocity of sound in air is 330 m/s. The beats frequency received by the observer is



(A) 2 Hz (B) 4 Hz
(C) 6 Hz (D) 8 Hz

16. Inside a uniformly charged infinitely long cylinder of radius 'R' and volume charge density 'ρ' there is a spherical cavity of radius 'R/2'. A point 'P' is located at a distance 2R from the axis of the cylinder as shown. Then the electric field strength at the point 'P' is



(A) $\frac{23\rho R}{54\varepsilon_0}$ (B) $\frac{23\rho R}{108\varepsilon_0}$

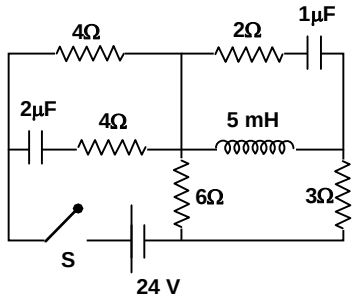
(C) $\frac{25\rho R}{54\varepsilon_0}$ (D) $\frac{25\rho R}{108\,\varepsilon_0}$

17. A parallel plate capacitor 'A' of capacitance 1 μF is charged to a potential drop 100 volt and then disconnected from the battery. Now the capacitor 'A' is completely filled with a dielectric slab of dielectric constant k = 4. Another parallel plate capacitor 'B' of capacitance 2 μF is charged to a potential drop 20 volt and then disconnected from the battery. Now the two capacitors 'A' and 'B' are connected with each other with opposite polarities. Then the heat dissipated after connecting the capacitors is

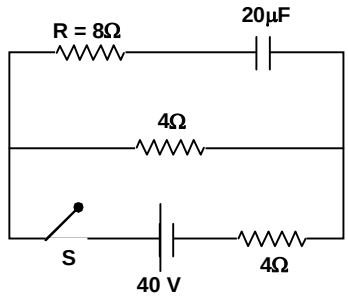
(A) 1.15×10^{-3} J (B) 1.25×10^{-3} J
(C) 1.35×10^{-3} J (D) 1.45×10^{-3} J

Space for rough work

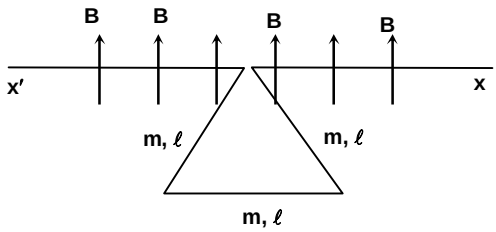
18. In the circuit shown, the switch 'S' is closed at t = 0. Then the current through the battery after steady state reached is
(A) 2A
(B) 4A
(C) 6A
(D) 8A



19. In the circuit shown, the switch 'S' is closed at t = 0. The capacitor is initially uncharged. Then the current through the resistor R = 8Ω at t = 0.4×10^{-3} sec is (Take $e^{-2} = 0.135$)
(A) 0.18 A
(B) 0.27 A
(C) 0.54 A
(D) 0.72 A



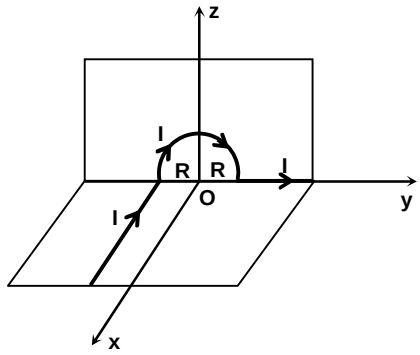
20. A uniform wire of mass '3m' and length '3l' is bent into the shape of an equilateral triangle and suspended from a horizontal axis X'X in a vertical plane as shown. A uniform vertical upward magnetic field 'B' is existing in the region. If charge 'Q' is passed almost instantaneously through the triangular loop, then the angular velocity acquired by the triangular loop immediately after charge passed is



(A) $\frac{\sqrt{3}BQ}{5m}$
(B) $\frac{2\sqrt{3}BQ}{5m}$
(C) $\frac{3\sqrt{3}BQ}{5m}$
(D) $\frac{4\sqrt{3}BQ}{5m}$

Space for rough work

21. A long wire carrying current 'I' is bent into the shape as shown in the figure. The net magnetic field intensity at the centre 'O' is



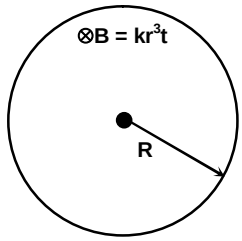
(A) $\frac{\mu_0 I}{4\pi R}\sqrt{1+\pi^2}$

(B) $\frac{\mu_0 I}{4\pi R}\sqrt{2+\pi^2}$

(C) $\frac{\mu_0 I}{4\pi R}\sqrt{4+\pi^2}$

(D) $\frac{\mu_0 I}{4\pi R}\sqrt{9+\pi^2}$

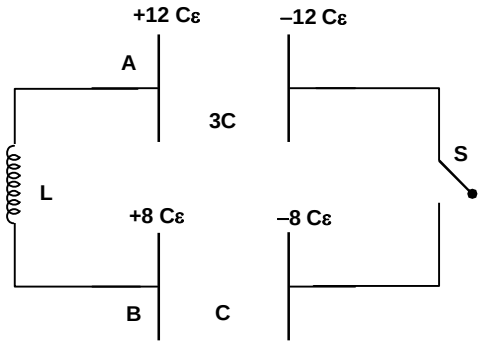
22. A time varying magnetic field $B = Kr^3t$ is existing in a cylindrical region of radius 'R' as shown (where k is a constant). The induced electric field 'E' at $r = \frac{R}{2}$ is



(A) $\frac{KR^4}{20}$ (B) $\frac{KR^4}{40}$

(C) $\frac{KR^4}{60}$ (D) $\frac{KR^4}{80}$

23. In the adjacent circuit, the capacitor 'A' of capacitance '3C' and the capacitor 'B' of capacitance 'C' are initially charged by potential drops 4ε and 8ε respectively and connected through an inductor of inductance 'L' with zero initial current as shown. The switch 'S' is closed at t = 0. then the maximum current through the inductor is



(A) $\varepsilon\sqrt{\frac{3C}{L}}$

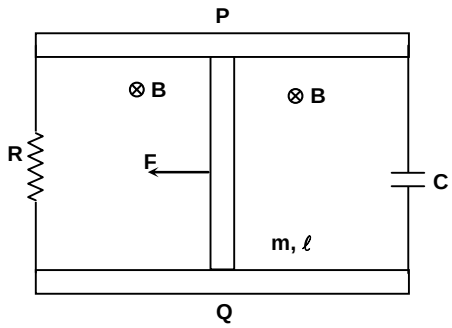
(B) $\varepsilon\sqrt{\frac{6C}{L}}$

(C) $\varepsilon\sqrt{\frac{12C}{L}}$

(D) $\varepsilon\sqrt{\frac{15C}{L}}$

Space for rough work

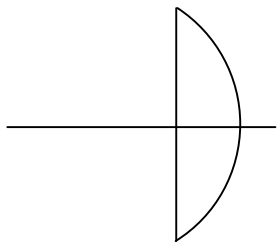
24. Two smooth horizontal parallel conducting rails are connected with a capacitor and a resistor at the two ends as shown. A uniform conducting rod PQ of mass 'm' and length 'ℓ' is dragged with a constant horizontal force 'F'. The terminal velocity acquired by the conducting rod is



(A) $\frac{4FR}{B^2\ell^2}$ (B) $\frac{2FR}{B^2\ell^2}$

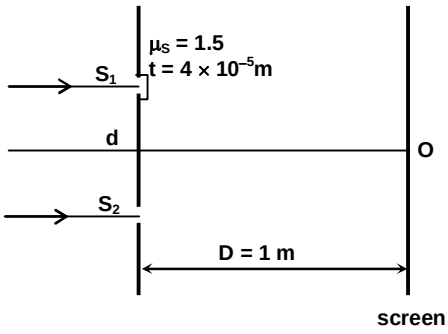
(C) $\frac{FR}{B^2\ell^2}$ (D) $\frac{FR}{2B^2\ell^2}$

25. A plano-convex lens $(\mu = 1.5)$ of aperture diameter 8 cm has a maximum thickness of 4 mm. The focal length of the lens relative to air is



(A) + 20 cm
(B) + 40 cm
(C) + 60 cm
(D) + 80 cm

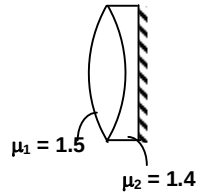
26. A coherent parallel beam of light of wavelength 5000 Å is incident normally on the plane of slits. The slit S_1 is covered with a slab of refractive index $\mu_s = 1.5$ and thickness $t = 4\times10^{-5}$m as shown. The separation between the slits is d = 1 mm and the separation between the plane of slits and screen is D = 1m. Then the position of the central maxima formed on the screen is



(A) 1 cm above point 'O'.
(B) 2 cm above point 'O'.
(C) 1 cm below point 'O'.
(D) 2 cm below point 'O'.

Space for rough work

27. A thin equiconvex glass lens $(\mu_1 = 1.5)$ of radius of curvature 'R' and a thin plano-concave lens $(\mu_2 = 1.4)$ are kept in contact and the plane surface is silvered. This lens system is placed in air. When a point object is placed at a distance 20 cm from the lens system on its principal axis, the image formed by the lens system coincides with the object. Then the value of radius of curvature 'R' is



(A) 12 cm (B) 18 cm
(C) 24 cm (D) 30 cm

28. When the light of wavelength 400 nm is incident on a metal surface of work function 2.3 eV, photoelectrons are emitted. A fastest photoelectron combines with a He^{2+} ion to form He^{+} ion in its third excited state and a photon is emitted in this process. Then the energy of the photon emitted during combination is
(A) 4.2 eV (B) 5.7 eV
(C) 6.5 eV (D) 6.8 eV

29. An unpolarised light of intensity 'I_0' is passed through the two Polaroid's placed parallel to each other and their transmission axes are inclined at an angle 45. The intensity of the polarised light after passing through the Polaroid's will be
(A) $\frac{I_0}{8}$ (B) $\frac{I_0}{4}$
(C) $\frac{I_0}{2}$ (D) $\frac{I_0}{\sqrt{2}}$

30. The pitch of a screw gauge is 1mm and its cap is divided into 100 divisions. When this screw gauge is used to measure the diameter of a wire, the main scale reading is 2mm and 58th division of circular scale coincides with the main scale. There is no zero error in the screw gauge. Then the diameter of the wire is
(A) 2.38 mm (B) 2.48 mm
(C) 2.58 mm (D) 3.58 mm

Space for rough work

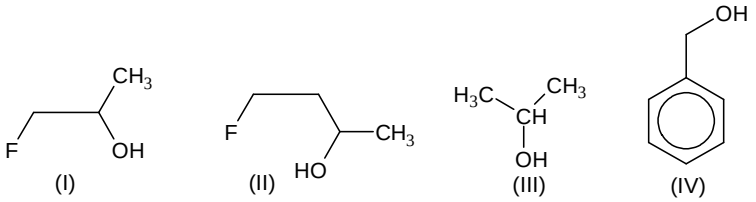
Chemistry PART – II

SECTION – A

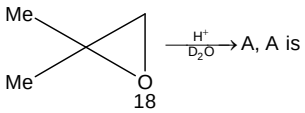
Straight Objective Type

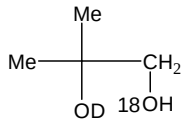
This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

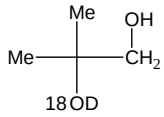
31. The order of reactivity of the following alcohols towards conc. HCl is

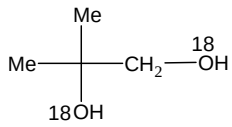


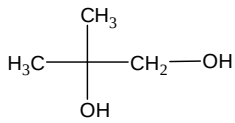
(A) IV > III > I > II
(B) IV > III > II > I
(C) I > III > II > IV
(D) I > II > III > IV

32. 

(A) 

(B) 

(C) 

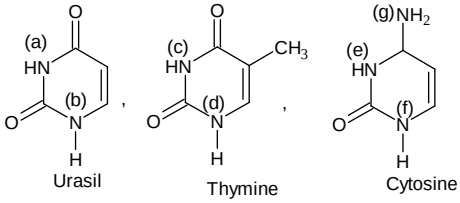
(D) 

33. The major product formed by the reaction of a anisol with tert-butyl alcohol in presence of Li/Liq. NH_3 is
(A) 1-methoxy cyclohexa-1,4-diene
(B) 2-methoxy cyclohexa-1,3-diene
(C) 1-methoxy cyclohexa-1, 3-diene
(D) 3-methoxy cyclohexa-1,4-diene

34. Consider the following statements for monosaccharides
(1) They are optically active polyhydroxy carbonyl compounds.
(2) Fructose is a ketose sugar and hence it does not give red precipitate with Fehling solution.
(3) All monosaccharides show mutarotation.
(4) α-D-Glucose and β-D-Glucose are anomers.
Which one of the above statement(s) is/are correct?
(A) 1 and 2
(B) 1, 2 and 3
(C) 1 and 3
(D) 1, 2 and 4

Space for rough work

35. Consider the following three bases of DNA and RNA:



The order of acidic strength on different site will be

(A) a > b
(B) d > c
(C) f > e > g
(D) g > f

36. The Arrhenious equation for the decomposition of methyl nitrite and ethyl nitride are k_1 and k_2 respectively and the activation energy of second is 4 kJ/mole is more than that of first. At what temperature value of rate constant k_1 will be equal to k_2 if

$$k_1(s) = 10^{13}\left(e^{-\frac{E_{a_1}}{RT}}\right) \text{and } k_2\left(s^{-1}\right) = 10^{14}\left(e^{-\frac{E_{a_2}}{RT}}\right)$$

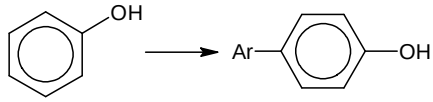
(A) 650 K
(B) 230 K
(C) 325 K
(D) 425 K

37. If the orbital angular momentum of iron chloride is $5\sqrt{6}\,\hbar$ then the correct options for this given fact are:

(A) $FeCl_3$
(B) $FeCl_2$
(C) Fe_2Cl_6
(D) None of these

38. Which of the following reactions are possible?

(1) $ArN_2^+ + CuBr \longrightarrow Ar - Br$

(2) $ArN_2^+ +$  $\longrightarrow$

(3) $ArN_2^+ + H_3PO_2 \xrightarrow{H_2O} ArH$

(4) $ArN_2^+ + I^- \longrightarrow ArI$

Select the answer using codes given below:

(A) 1, 2 & 4
(B) 1, 3 & 4
(C) 1, 2 & 3
(D) 1, 2, 3 & 4

39. The number of β-particle emitted during the change ${}_aX^c \longrightarrow {}_dY^b$ is

(A) $\frac{a-b}{4}$
(B) $d+\left(\frac{a-b}{2}\right)+c$
(C) $d+\left(\frac{c-b}{2}\right)-a$
(D) $d+\left(\frac{a-b}{2}\right)-c$

Space for rough work

40. In which type of voids magnesium and silicon is present respectively in asbestos ($Mg_3Si_2O_5$) if

$r_{Mg^{2+}} / r_{O^{2-}} = 0.59$ and $\frac{r_{Si^{4+}}}{r_{O^{2-}}} = 0.29$

(A) octahedral and tetrahedral
(B) Tetrahedral and octahedral
(C) both in tetrahedral
(D) both in octahedral

41. Ferric ion forms a Prussian blue coloured precipitate due to formation of
(A) $KMnO_4$
(B) $K_4[Fe(CN)_6]$
(C) $Fe_4[Fe(CN)_6]_3$
(D) $[Fe(OH)_3]$

42. Arseneous sulphide is a negative sol. The reagent with least coagulation power in this sol is
(A) $AlCl_3$
(B) NaCl
(C) CaF_2
(D) Glucose

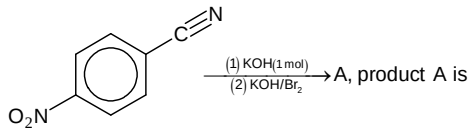
43. A 3 mole sample of a triatomic ideal gas at 300 K is allowed to expand under adiabatic reversible condition from 5 L to 40 L. The value of ΔH is
(A) - 12.46 kJ
(B) - 14. 965 kJ
(C) - 24.62 kJ
(D) - 10.24 kJ

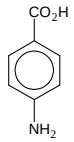
44. An aq. solution of a salt (x) is alkaline to litmus. On strong heating, it swell up to give a glassy material. When conc. H_2SO_4 is added to a concentrated solution of x, white crystal of a weak acid separates out. Hence compound (x) is
(A) $Na_2SO_4.10H_2O$
(B) $MgSO_4.7H_2O$
(C) $Na_2B_6O_{11}$
(D) $Na_2B_4O_7.10H_2O$

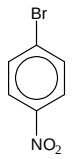
45. A gas is heated in such a way that its pressure and volume both become double. Now by decreasing temperature, some of air molecules were introduced into the container to maintain the increased volume and pressure. Assuming $1/4^{th}$ of the initial number of moles has been given for this purpose. By what fraction of temperature has been raised finally of initial absolute temperature.
(A) $\frac{4}{5}$ times
(B) $\frac{16}{7}$ times
(C) $\frac{16}{5}$ times
(D) $\frac{8}{5}$ times

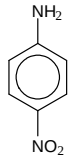
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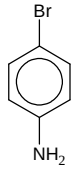
46. During the reduction of Gold(III) from $AuCl_4^-$ ion, the solution must be stirred rapidly during electrolysis. The stirring is important because
(A) It helps to separate gold from chlorine
(B) It increases the movement of $AuCl_4^-$ towards anode
(C) It overcomes the repulsion between the $AuCl_4^-$ and the excess e^- at the electrode and help in deposition of gold (III)
(D) It increases the conductivity of the solution.

47. 

(A) 

(B) 

(C) 

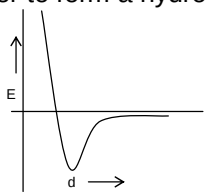
(D) 

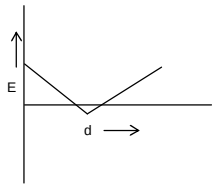
48. A well is dry in a bed of rock containing fluorspar (CaF_2). If the well contains 20,000 L of H_2O, what is the amount of F^- in it
K_{SP} of $CaF_2 = 4 \times 10^{-11}$.
(A) 4.3 mole (B) 6.8 mole
(C) 8.6 mole (D) 13.6 mole

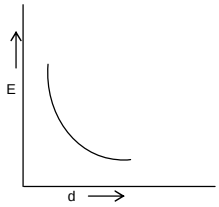
49. The electronaffinity of F < Cl and ionization energy of F > Cl then which of the compound is possible?
(A) $\overset{\delta+}{F}-\overset{\delta-}{Cl}$ (B) $\overset{\delta+}{Cl}\overset{\delta-}{F}$
(C) F – Cl (D) Cl – F_2

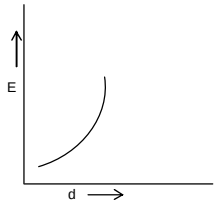
Space for rough work

50. Which plot best represent the potential energy (E) of two hydrogen atoms as they approach one another to form a hydrogen molecule?

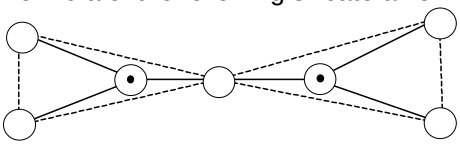
(A) 

(B) 

(C) 

(D) 

51. Formula of the following silicate anion is



(A) SiO_4^{4-} (B) $Si_2O_7^{6-}$

(C) $Si_2O_8^{6-}$ (D) $Si_3O_9^{6-}$

52. S – S bond is present in

(A) α - $(SO_3)_n$ (B) γ - (S_3O_9)

(C) $H_2S_2O_3$ (D) $H_2S_2O_8$

53. Main source of lead is PbS. It is converted to Pb by

(I) $PbS \xrightarrow[\Delta]{air} PbO + SO_2$

(II) $PbS \xrightarrow{air} PbO + PbS$, $PbS \rightarrow Pb + SO_2$

Self reduction process is

(A) I (B) II

(C) Both A & B (D) none of these

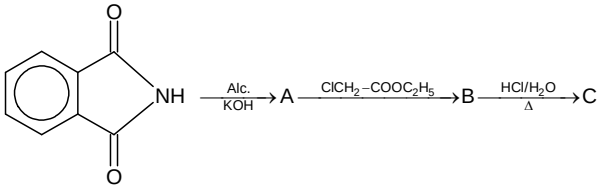
54. How many moles of HCl will be required to completely convert 1 moles of borax into boric acid?

(A) 1 (B) 2

(C) 3 (D) none of these

Space for rough work

55. Consider the following sequence of reaction:

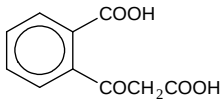


Which of the following is the structure of 'C'?

(A) $Cl^{-}\overset{+}{N}H_3CH_2-COOH$

(B) $NH_2 - CH_2 - COCl$

(C) $C_6H_5CONH_2$

(D) 

56. Consider the following sequence of reactions:

$HC \equiv CH \xrightarrow[2.\,HCl]{1.\,CuCl.NH_4Cl} A \xrightarrow{\text{Polymerisation}} B$

The polymer 'B' obtained is:

(A) Neoprene

(B) Chloroprene

(C) Polyvinyl chloride

(D) Orlon

57. 50 ml of 1 M, oxalic acid is shaken with 0.6 gm of charcoal. The final concentration of this solution after adsorption has been found to be 0.4 M. The amount of oxalic acid absorbed per gm of charcoal is

(A) 2.02 gm

(B) 2.52 gm

(C) 3.02 gm

(D) 3.52 gm

58. $Na_2B_4O_7.\ 10H_2O$ (Borax) is correctly represented as

(A) $2Na.BO_2.\ Na_2B_2O_2.\ 10H_2O$

(B) $Na_2[B_4O_5(OH)_4].8H_2O$

(C) $Na_2[B_4(H_2O)_4].6H_2O$

(D) all of the above.

Space for rough work

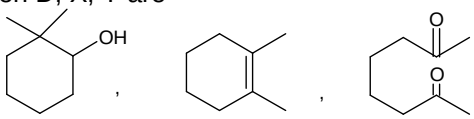
59. A compound 'D' ($C_8H_{16}O$) has following observation

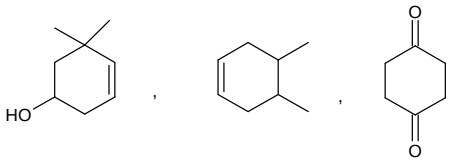
(a) It reacts with Lucas reagent with in 5 min.

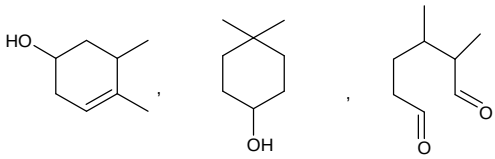
(b) On heating in presence of acid gives a compound X, which can decolorize $KMnO_4$ as well as Br_2/CCl_4.

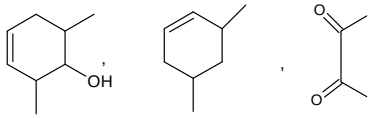
(c) Compound X $\xrightarrow{\text{ozonolysis}}$ Y $\xrightarrow{I_2/NaOH}$ 2 moles of iodoform + sodium salt of dibasic acid $\xrightarrow{\text{Sodalime}}$ n – butane

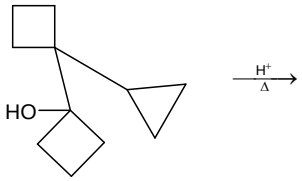
Then D, X, Y are

(A) 

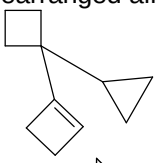
(B) 

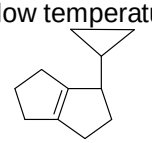
(C) 

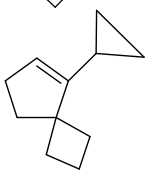
(D) 

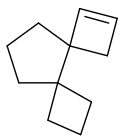
60. 

The rearranged alkene, product after rearrangement at low temperature will be mainly:

(A) 

(B) 

(C) 

(D) 

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains **30 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

61. $\int \frac{(\log x)^2 \tan^{-1}(\log x) + 1 + \tan^{-1}(\log x)}{1+(\log x)^2} dx$

(A) $\frac{x}{1+(\log x)^2} + c$ (B) $\frac{x}{\tan^{-1}(\log x)} + c$

(C) $x \tan^{-1}(\log x) + c$ (D) $\log x(\tan^{-1}(\log x)) + c$

62. $\int \frac{11x-18}{x^{11}\left(\sqrt[6]{x^{12}-2x+3}\right)} dx$ is equal to

(A) $\frac{6}{5}\frac{\left(x^{12}-2x+3\right)^{\frac{5}{6}}}{x^{10}}$ (B) $\frac{3}{5}\frac{\left(x^{12}-2x+3\right)^{\frac{5}{6}}}{x^{10}} + c$

(C) $\frac{3}{5}\left(x^{12}-2x+3\right)^{\frac{5}{6}} + c$ (D) $\frac{5}{3}\frac{\left(x^{12}-2x+3\right)}{x^2} + c$

63. Let $\frac{d}{dx}(g(x)) = \frac{e^{\cos x}}{x}$, x > 0, g(1) = $3a_1$, g(2) = $6a_2$, g(8) = $10a_3$ (where a_1 a_2, a_3 are constants) then $\int_1^2 \frac{3e^{\cos x^3}}{x} dx$ is equal to

(A) $6a_2 - 3a_1$ (B) $10a_3 - 6a_2$

(C) $10a_3 - 3a_1$ (D) $10a_3 - 6a_2 - 3a_1$

Space for rough work

64. A circle with centre in the first quadrant has tangent as y = x + 10, y = x – 6 and y-axis. Let (h, k) be the centre of the circle. If $h+k=a+b\sqrt{a}$, where (a, b $\in$ Q), then a + b is equal to
(A) 8 (B) 10
(C) 12 (D) 0

65. If we add 1, 1, 7 to first three terms of an increasing A.P. whose common difference is an integer, then the resulting three terms are in an increasing G.P. whose common ratio is also an integer, then the number of possible values of d are
(A) 2 (B) 4
(C) 8 (D) infinite

66. If the line $\frac{x-2}{3}=\frac{y+1}{2}=\frac{z-3}{4}$ lies in the plane ax + by + 3z = 8, then $a^2 + b^2$ is equal to
(A) 16 (B) 8
(C) 13 (D) 25

67. If the curve y = f(x) passes through (1, 1) and satisfies the differential equation $2x^2y\,dx - x^3\,dy = y^2\,dx$, then f(2) is equal to
(A) $\frac{2}{(\log 2)+1}$ (B) $\frac{4}{\log 2}$
(C) $\frac{4}{\log 2e}$ (D) $\frac{16}{(\log 2)+1}$

68. If $\frac{1}{\sqrt{\alpha}}$ and $\frac{1}{\sqrt{\beta}}$ are the roots of the equation $px^2 + qx + 1 = 0$ ($p \neq 0$, p, q $\in$ R) then the equation $x(x + q^3) + (p^3 - 3pqx) = 0$ has roots
(A) $\alpha^{\frac{3}{2}}$ and $\beta^{\frac{3}{2}}$ (B) $\alpha\beta^{\frac{1}{2}}$ and $\alpha^{\frac{1}{2}}\beta$
(C) $\sqrt{\alpha\beta}$ and $\alpha\beta$ (D) $(\alpha\beta)^{\frac{1}{4}}$ and $(\alpha\beta)^{\frac{5}{4}}$

Space for rough work

69. If f(x) is continuous and $f\left(\frac{25}{2}\right) = 4$ then $\lim_{x\to 0} f\left(\frac{1-\cos 5x}{x^2}\right)$ is equal to

(A) $\frac{1}{4}$ (B) 0

(C) 8 (D) 4

70. Let A = {(x, y) : $y^2 \le 8x$, $3y - 4x \ge 4$}. The area (in square units) of the region A is

(A) $\frac{1}{3}$ (B) $\frac{1}{6}$

(C) 1 (D) 2

71. If the point (2, 3) lies inside the circle $x^2 + y^2 - 6x - 10y + k = 0$ and the circle neither touches nor intersect the line y = 1 then the set of all possible value of k is in the interval

(A) no values of k (B) (9, 29)

(C) (18, 29) (D) (25, 31)

72. In a triangle ABC, angle C < angle B, angle B and angle C satisfy the equation $2\tan x - \alpha(1 + \tan^2 x) = 0$ (where $0 < \alpha < 1$). Then the measure of angle A is

(A) $\frac{\pi}{3}$ (B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

73. If A is a 3×3 matrix such that $A^T = 5A + 2I$, where A^T is the transpose of A and I is the 3×3 identity matrix, then there exist a column matrix $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ then AX is equal to

(A) AX = X (B) $AX = -\frac{X}{2}$

(C) AX = –2X (D) AX = 0

74. The contrapositive of the converse of $\sim q \to p$ is

(A) $q \to \sim p$ (B) $p \to \sim q$

(C) $\sim p \to q$ (D) none of these

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75. The locus of the mid-point of the chords of the ellipse $49x^2 + 16y^2 = 784$, the tangents at the ends of which intersect on the circle $x^2 + y^2 = 100$ is

(A) $\left(49x^2+16y^2\right)^2=\left(\frac{784}{10}\right)^2\left(x^2-y^2\right)$ (B) $49x^2+16y^2=\frac{784}{10}$

(C) $\left(49x^2+16y^2\right)^2=\frac{784^2}{10^2}\left(x^2+y^2\right)$ (D) none of these

76. Box A contains four tickets numbered 1, 2, 3, 4, box B contains 7 tickets numbered 1, 2, 3, 4, 5, 6, 7. A ticket is drawn from each of the boxes. Let a_i be the number on the ticket drawn from Box A and let b_i be the number on the ticket drawn from box B. The probability that $a_i \times b_i$ is a multiple of 3 is

(A) $\frac{15}{28}$ (B) $\frac{13}{28}$

(C) $\frac{2}{3}$ (D) $\frac{11}{28}$

77. The sum $\sum_{r=0}^{30} {}^{30}C_r \sin rx \cos(30-r)x$ is equal to

(A) $2^{30}\cos 30x$ (B) $2^{29}\cos 30x$

(C) $2^{29}\sin 29x$ (D) $2^{29}\sin 30x$

78. Let A and B be two sets of two complex numbers defined as follows $A = \{z : |z + 6| + |z - 2| = 10\}$, $B = \{z : |z - 6| = 3\}$ then the number of elements common to both A and B is

(A) 1 (B) 2

(C) 0 (D) 4

79. If the tangent to the conic $y - 4 = x^2$ at (3, 13) touches the circle $x^2 + y^2 - 4x + 2y + \lambda = 0$ (for some fixed λ) at a point (a, b), then a + b is equal to

(A) $\frac{55}{37}$ (B) $\frac{3}{37}$

(C) $-\frac{55}{37}$ (D) $-\frac{3}{37}$

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80. The value of $\lim\limits_{n\to\infty}\sum\limits_{r=n^2}^{n^2+9+6n}\frac{1}{\sqrt{r}}$ is equal to

(A) 6 (B) 4

(C) $-2\sqrt{2}$ (D) cannot be determined

81. Let f: R $\to$ R be a continuous and differentiable function such f(x) = f(10 – x) and slope of tangent at x = –5 is 2 then f′(5) + f′(15) is equal to

(A) 0 (B) 2

(C) –2 (D) –4

82. Let f(2) = –5 and f′(x) $\geq$ 5.4 for $2 \leq x \leq 7$. Then possible value of f(7) lies in the interval

(A) $(-\infty, 21]$ (B) [12, 18]

(C) (0, 22) (D) $[22, \infty)$

83. The value of $\sum\limits_{k=1}^{13}\tan\frac{k\pi}{12}\tan\frac{(k-1)\pi}{12}$ is

(A) 0 (B) –14

(C) –12 (D) $\frac{\sqrt{3}+1}{\sqrt{3}-3}-13$

84. Let a and b $\in$ R and f: R $\to$ R be defined by $f(x)=ae^{|x|\sin|x|}+b\cos(|x|)$. Then which of the following is INCORRECT?

(A) differentiable at x = 0 if a = 1 and b = 0 (B) differentiable at x = π if a = 1 and b = 1

(C) non differentiable at x = 0 if a = 1 and b = 1 (D) differentiable at x = 2π if a = 1 and b $\in$ R

85. Let $\vec{a}=a_1\hat{i}+a_2\hat{j}+a_3\hat{k}$ be a unit vector in R^3 and $\vec{b}=2\hat{i}+3\hat{j}+6\hat{k}$. Given that there exists a vector $\vec{c}$ in R^3 such that $|\vec{a}\times\vec{c}|=5$ and $\vec{b}\cdot(\vec{a}\times\vec{c})=35$. Which of the following statement(s) is correct?

(A) if a lies in yz plane then $|a_2| = |3a_3|$ (B) if a lies in xy plane, then $|a_1| = \left|\frac{3}{2}a_2\right|$

(C) if a lies in xz plane then $|a_1| = |2a_3|$ (D) $\vec{b}$ is always parallel to $\vec{c}$

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86. Let a, b $\in$ R consider the system of linear equations 2x + ay + 6z = 8, x + 2y + bz = 5 and x + y + 3z = 4. Then which of the following statements is INCORRECT?
(A) if a = 2, then system of equations have infinitely many solution
(B) if a = 5 and b = 3 then the system of equations have no solution
(C) if a $\neq$ 2 and b = 3, then the system of equations have infinitely many solution
(D) if a $\neq$ 2 and b $\neq$ 3, then the system of equations have a unique solution

87. If the mean deviation about the median of the numbers k, 3k, 7k, 9k, 11k, 13k is 11 then k is equal to (k > 0)
(A) $\frac{5}{3}$ (B) $\frac{22}{7}$
(C) 3 (D) 6

88. If e_1, e_2, e_3, e_4, e_5 are eccentricities of the conics xy = 16, $x^2 - y^2 = 36$, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $\frac{x^2}{a^2} - \frac{y^2}{b^2} = -1$ and $y^2 = 32x$ and $\sqrt{\frac{1}{e_1^2} + \frac{1}{e_2^2} + \frac{1}{e_3^2} + \frac{1}{e_4^2} + \frac{1}{e_5^2}} = \tan\theta\left(0 < \theta < \frac{\pi}{2}\right)$ then θ is equal to
(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{12}$
(C) $\frac{\pi}{3}$ (D) $\tan^{-1}\left(\sqrt{5}\right)$

89. An eight digit made up of all distinct digits 5, 7, 2, 4, 9, a, b, c is divisible by 3. Then the possible number of ordered triplets (a, b, c) is
(A) 12 (B) 24
(C) 1840 (D) 18

90. Let f: (–1, 1) $\rightarrow$ R be a continuous and differentiable function. If $\int_0^{\sin x} f(t)dt = x^2$ then $f\left(\frac{1}{2}\right)$ is equal to
(A) $\frac{2\pi}{3}$ (B) $\frac{\pi}{3}$
(C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{3\sqrt{3}}$

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